Algorithm 1 Rapidly-exploring Random Tree (RRT), user-chosen samples

function RRT($q_{\text{start}}, q_{\text{goal}}, \Delta$)
  $T \leftarrow \{q_{\text{start}}\}$
  repeat
    $q_{\text{rand}} \leftarrow$ PointClickedByUser
    $q_{\text{near}} \leftarrow$ Nearest(T, q_{rand})
    $q_{\text{new}} \leftarrow$ Steer($q_{\text{near}}, q_{\text{rand}}, \Delta$)
    if $\neg$CollisionFree($q_{\text{near}}, q_{\text{new}}$) **then**
      continue
    end if
    $T \leftarrow T \cup \{q_{\text{new}}\}$
    if $\|q_{\text{new}} - q_{\text{goal}}\| \leq \Delta$ and CollisionFree($q_{\text{new}}, q_{\text{goal}}$) **then**
      $T \leftarrow T \cup \{q_{\text{goal}}\}$
      return ReconstructPath(T, parent, q_{goal})
    end if
  until false
end function